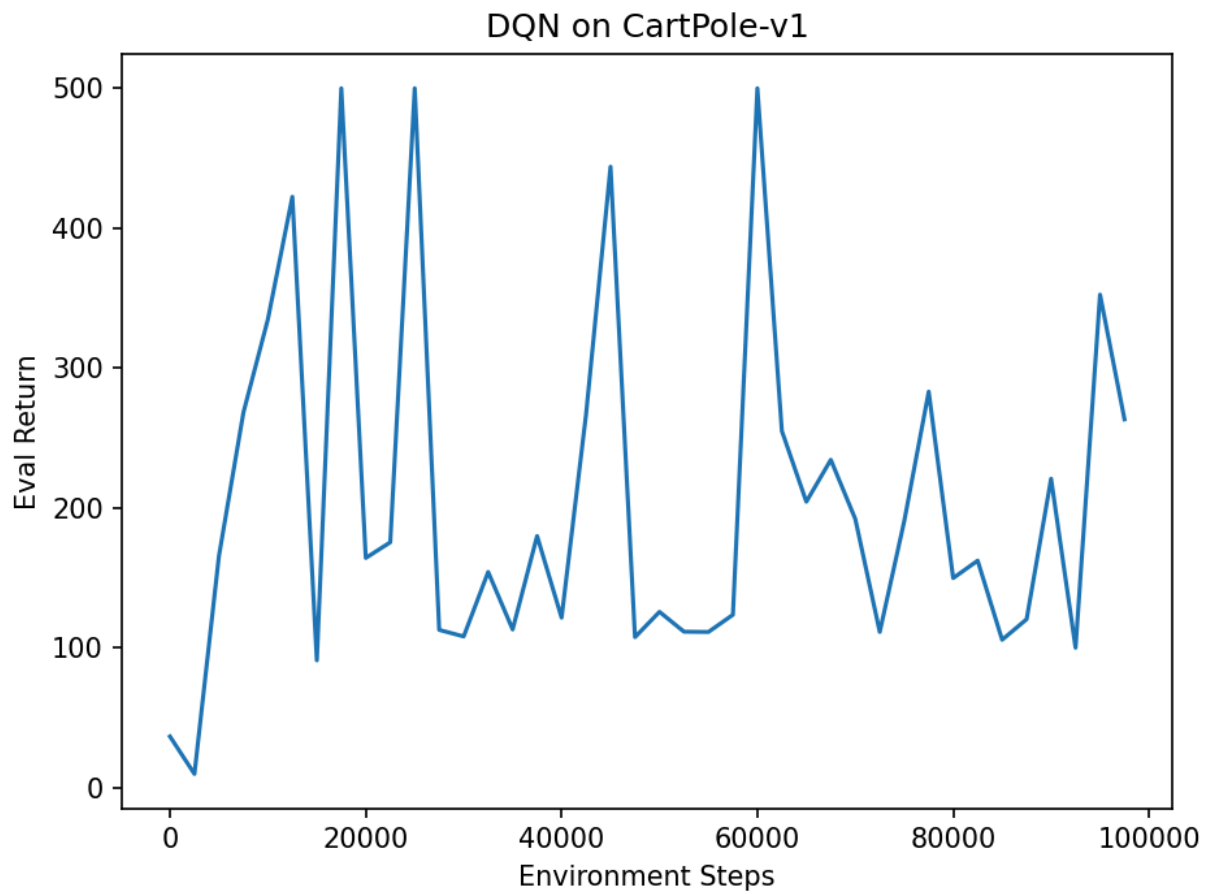


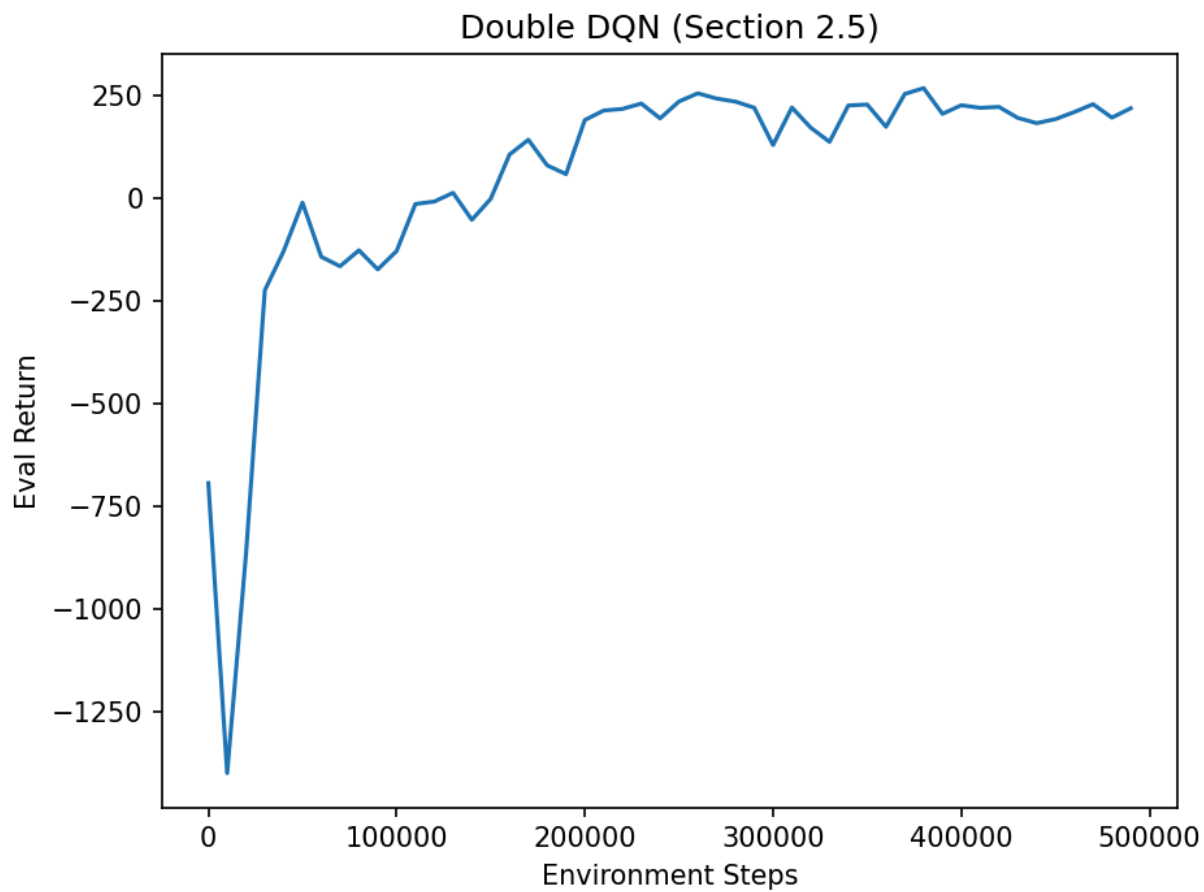
2.4 Basic Q-Learning

- CartPole-v1

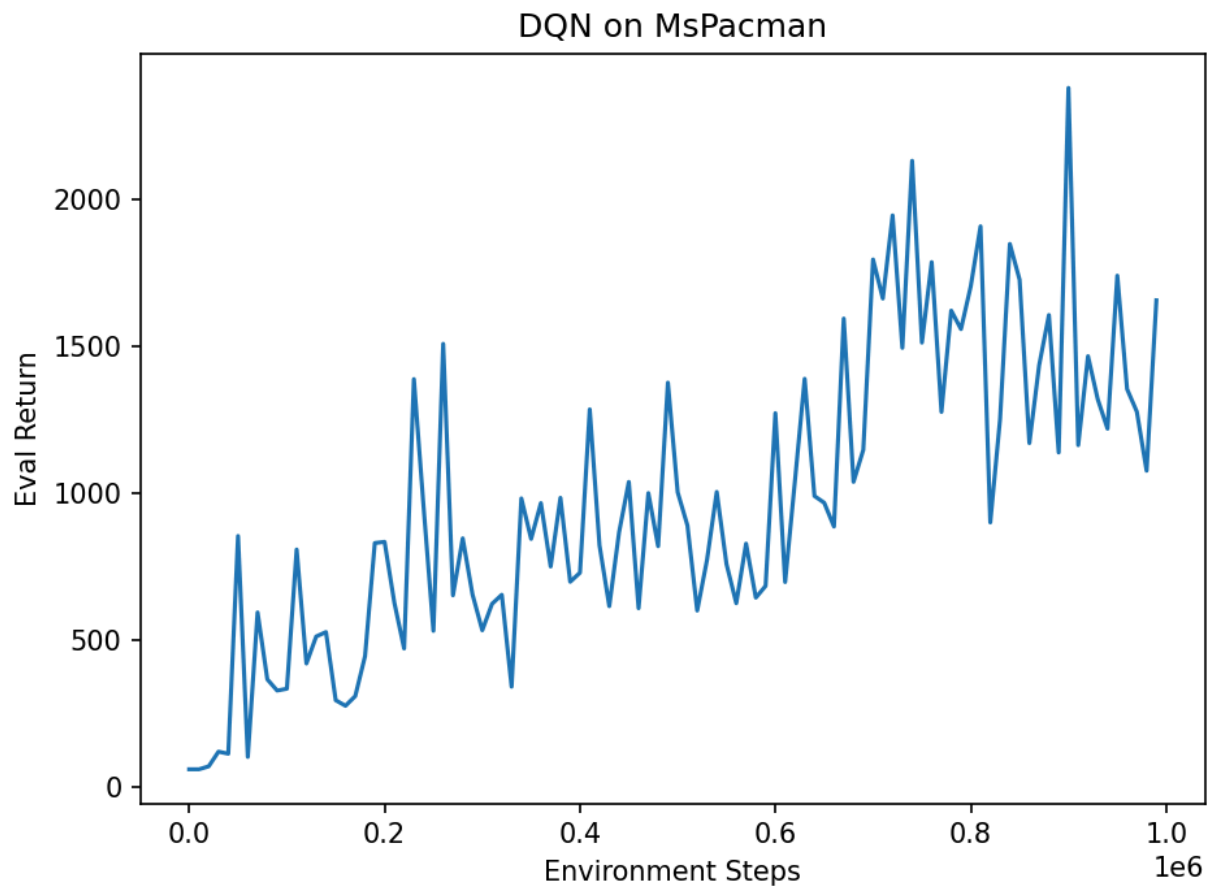


2.5 Double Q-Learning

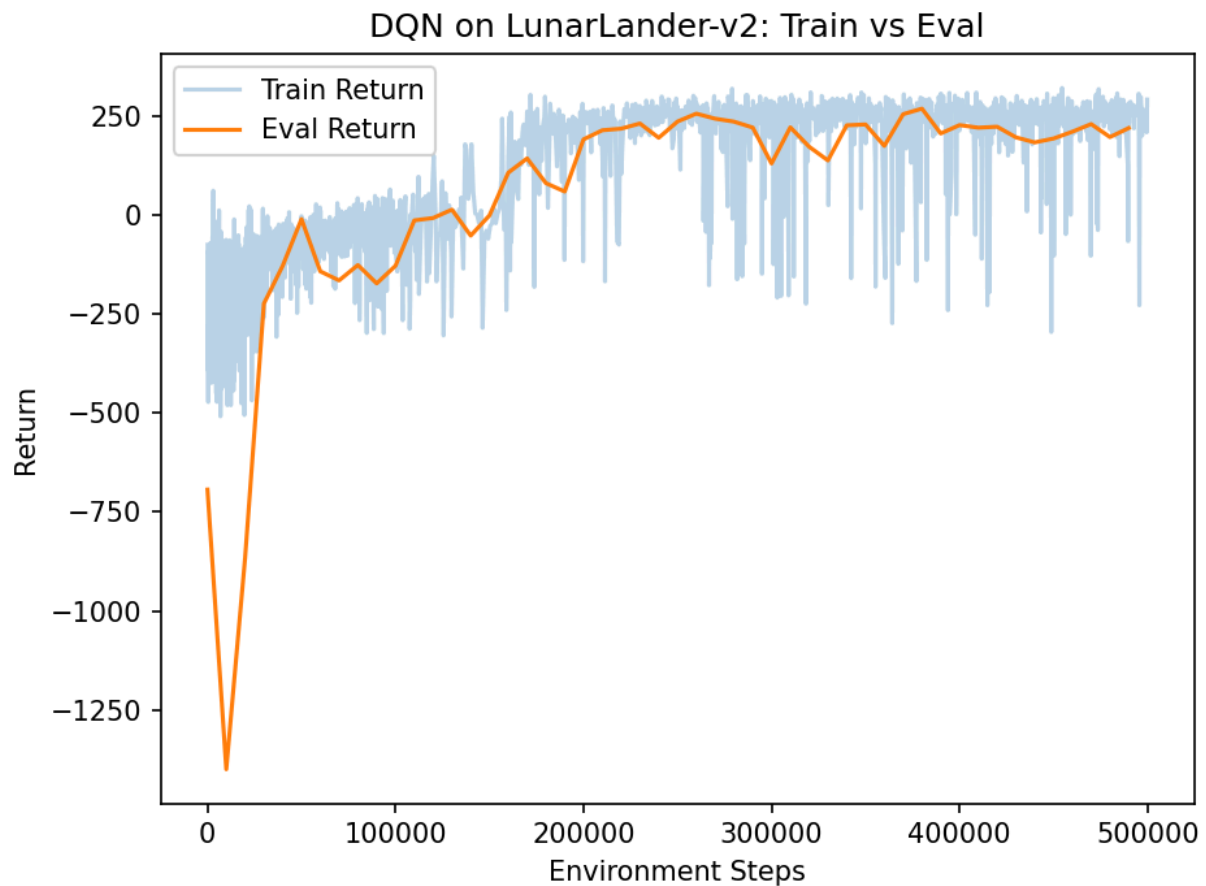
- LunarLander-v2



- MsPacman-v0

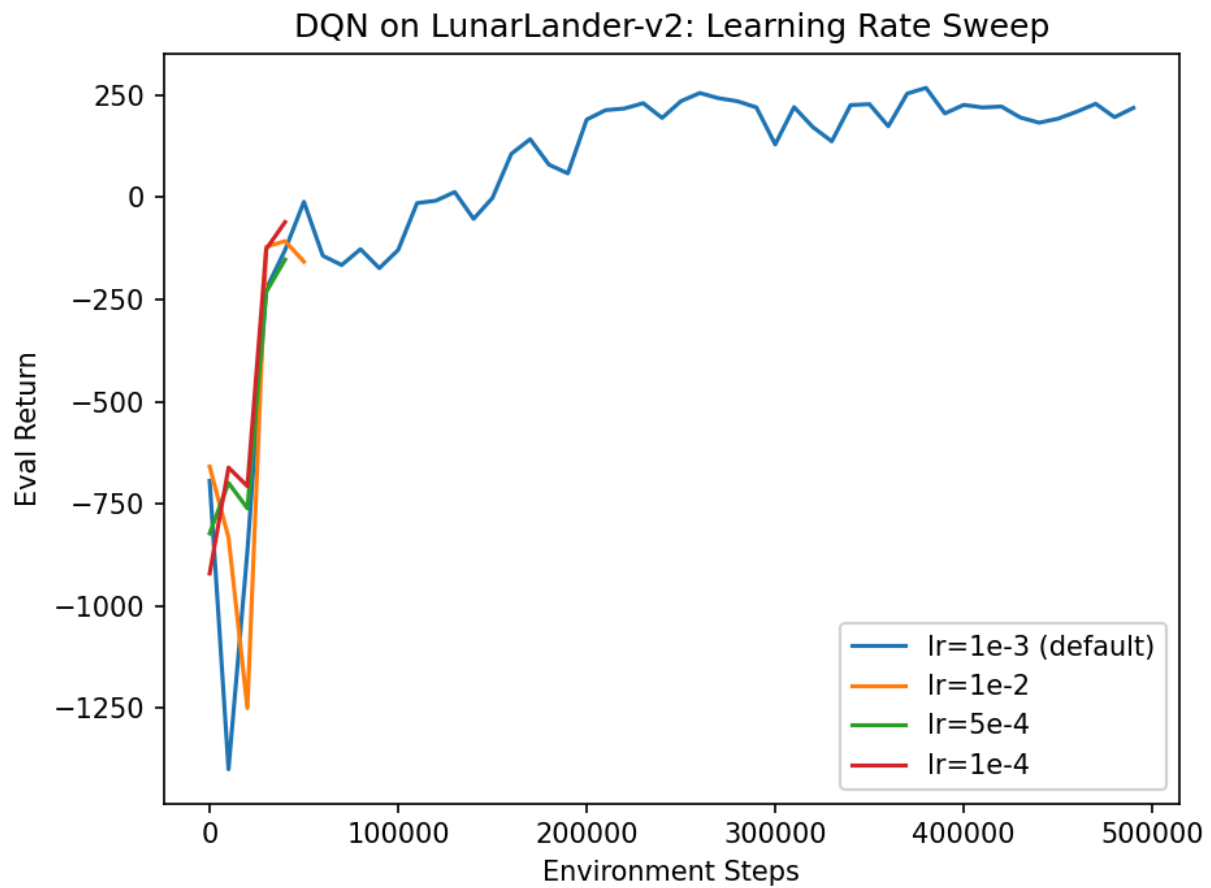


- Eval and train return on same axes



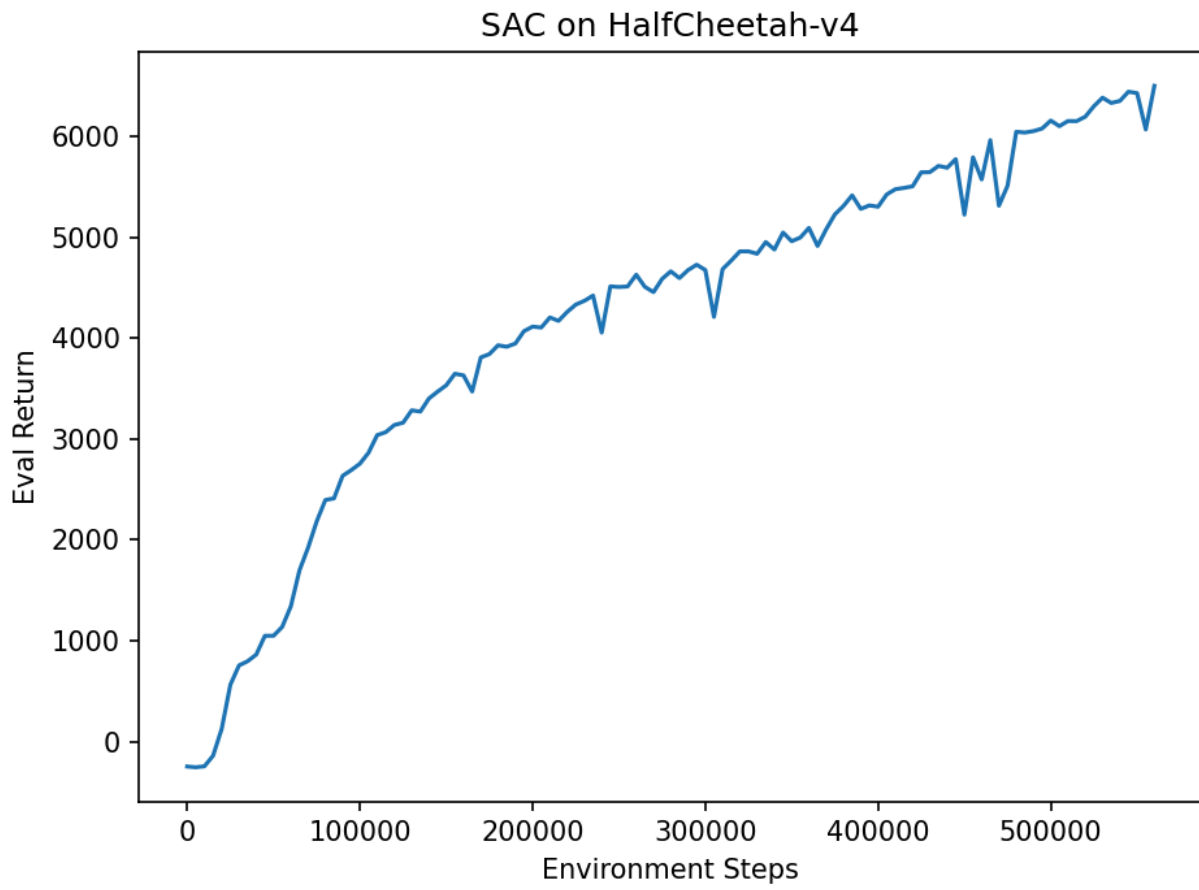
During training, the agent takes a lot of random actions because of the high epsilon value in epsilon-greedy exploration, which tanks the training returns early on. The eval returns are higher because evaluation uses a fully greedy policy with no random actions, so it actually uses what the agent has learned.

2.6 Experimenting with Hyperparameters

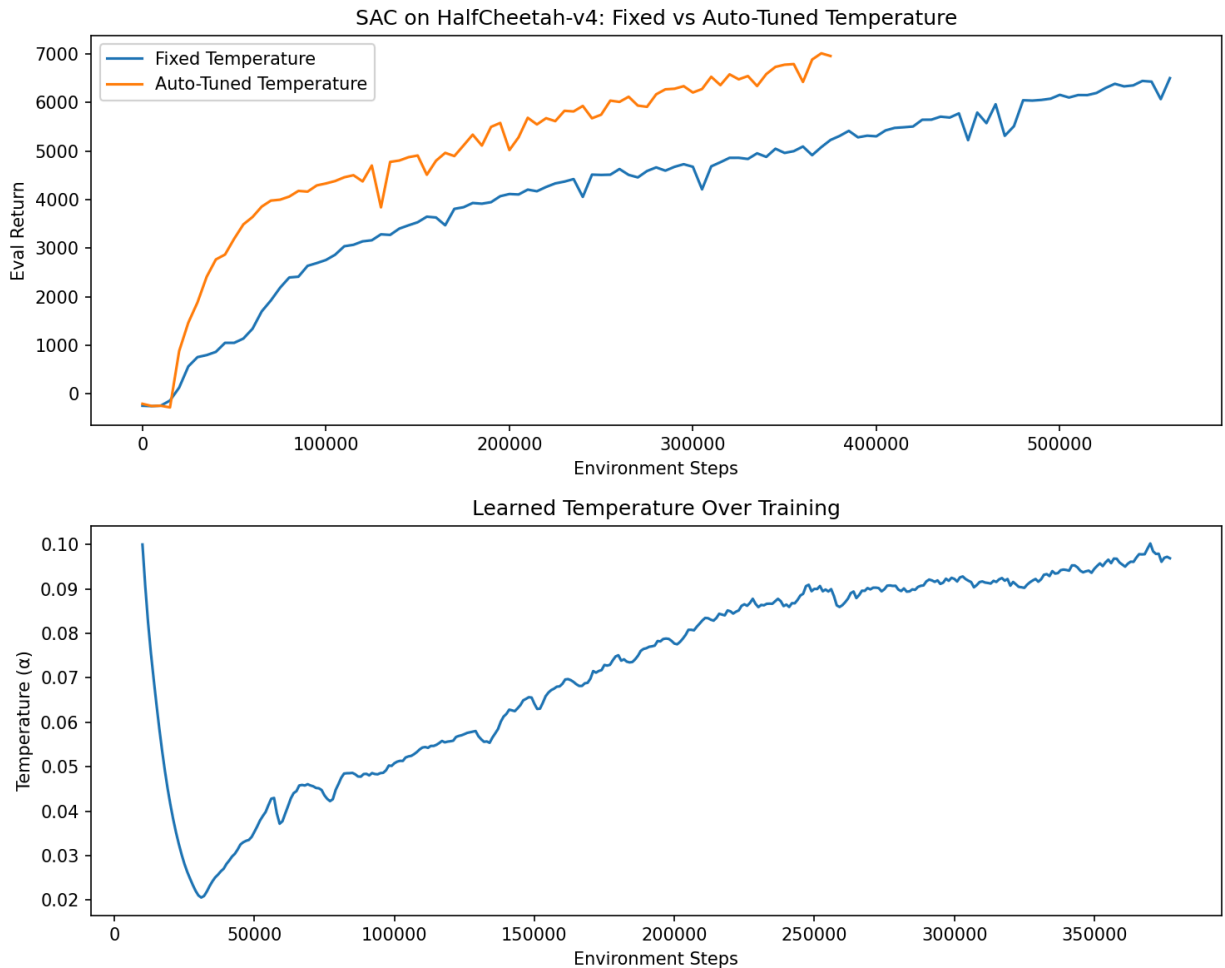


Chose learning rate because it's a good gauge of training dynamics

3.4 Actor Update with Reparametrization



3.5 Automatic Temperature Tuning



- Does auto-tuning match fixed?

Auto-tuning achieves comparable performance to the fixed temperature since the default fixed value was already well-tuned for HalfCheetah. The main advantage is not needing to hand-pick that value.

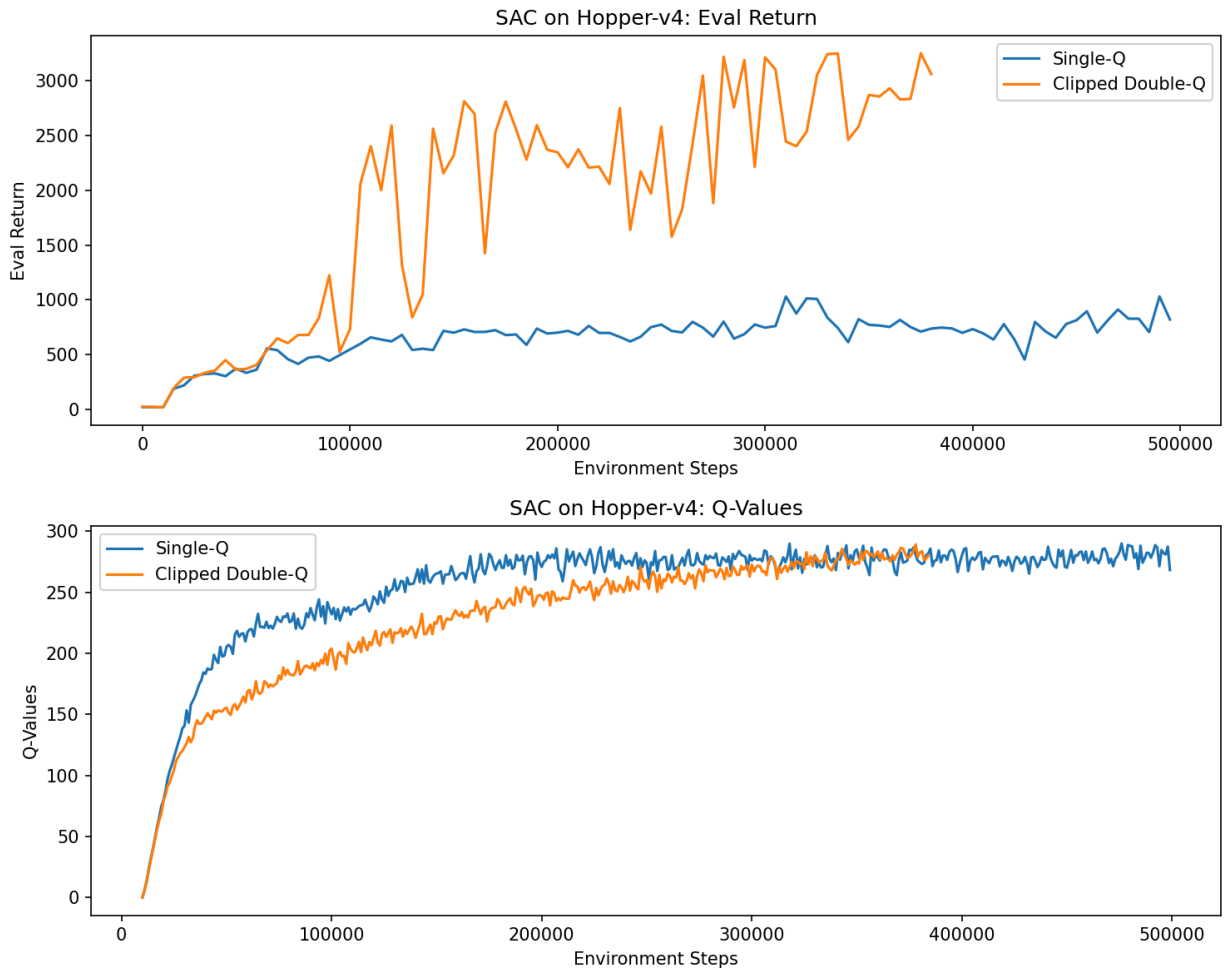
- How does temperature evolve?

The temperature typically decreases over training and then stabilizes at a lower value.

- Why?

Early in training the policy is bad and needs more exploration, so a higher temperature helps. As the agent learns a good policy for HalfCheetah's locomotion task, it becomes more confident about which actions to take, so the temperature drops to exploit the learned behavior instead of exploring randomly.

3.6 Stabilizing Target Values



Clipped double-Q works better. With single-Q, the critic overestimates Q-values because errors compound through bootstrapping the critic trains on its own noisy predictions, so positive errors accumulate and blow up. You can see this in the Q-values plot where single-Q has much higher (inflated) Q-values. These bad Q-values mislead the actor into picking actions that aren't actually good, hurting performance. Clipped double-Q fixes this by taking the minimum of two independent critics as the target, which keeps the Q-values grounded and leads to a more stable, better-performing policy